

Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024: A Source-Audited Panel Analysis

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Abstract

Permitless concealed carry laws allow eligible adults to carry concealed handguns in public without first obtaining a carry permit. The rapid spread of these laws creates a timely policy question, but estimation is difficult because adoption is politically patterned, legal classifications contain edge cases, and permitless-carry statutes often remove several pre-carry requirements at once. We assembled a 50-state annual panel from 1999 through 2024 using CDC mortality outcomes and a source-audited permitless-carry treatment file. Firearm suicide was the prespecified primary outcome; total suicide, non-firearm suicide, firearm homicide, and total firearm deaths were secondary outcome-family checks. In the baseline fixed-effects model with state and year fixed effects, unemployment, income, and state-clustered standard errors, permitless carry adoption was associated with a 1.263 deaths per 100,000 increase in firearm suicide rates (SE 0.227; $p < 0.001$). The estimate remained positive in population-weighted models (0.853; $p < 0.001$), pre-COVID and COVID-excluded samples, and staggered-adoption summaries based on cohort windows and not-yet-treated comparisons. It attenuated with state-specific linear trends (0.253; $p = 0.069$), and event-time estimates showed statistically significant pre-adoption signal. Firearm homicide showed no comparable baseline association (-0.072; SE 0.368; $p = 0.846$), although homicide-rate missingness was concentrated in the smallest-population state-years. Overall, permitless carry adoption was most consistently associated with firearm suicide and total suicide, not firearm homicide. The findings support a state-level observational association and should not be interpreted as individual-level causal evidence.

Introduction

Permitless concealed carry is a newer policy margin in a long-running concealed-carry debate. Earlier right-to-carry studies focused mostly on transitions from no-issue or may-issue systems to shall-issue permitting, asking whether broader legal carrying deters violent crime or increases firearm violence [1–3]. That literature remains methodologically contested, with replication work emphasizing sensitivity to event-study validity, heterogeneous treatment effects, and influential observations [4]. The policy landscape has since shifted again. Since 2015, many states have moved beyond shall-issue permitting by allowing eligible adults to carry concealed handguns without first obtaining a carry permit, a change that became still more salient after *Bruen* altered the constitutional environment for restrictive carry permitting [5]. The relevant empirical question is no longer only whether permits should be issued more easily, but what happens when the permit requirement is removed.

That distinction is substantively important. A carry-permit requirement can screen applicants, impose delay, require training, or document who is legally authorized to carry. A permitless-carry statute may remove a training requirement, a permit-background-check step, a permit-specific misdemeanor screen, or a discretionary suitability review, while leaving general firearm-possession prohibitions in place. Prior concealed-carry work has emphasized that these permit provisions can matter for firearm violence outcomes [2]. A recent permitless-carry working paper estimates increases in violent crime and firearm aggravated assault after adoption, reinforcing the need to evaluate permitless carry separately from older shall-issue transitions [6]. Recent firearm-suicide work has also reported that concealed-carry permit requirements are associated with lower firearm suicide rates, alongside handgun purchase permits and waiting periods [7]. The policy margin is therefore relevant to both suicide and homicide, but the pathway is not obvious *ex ante*.

Suicide and homicide also require separate analysis. Firearm access is strongly associated with firearm suicide at household, individual, and state levels [8–10]. Purchaser-licensing evidence links stronger handgun screening systems with lower firearm homicide and suicide in several state settings [11]. A law governing public carry may not directly change household gun ownership, but it can change the carrying environment, the perceived normality of immediate firearm availability, and the screening requirements attached to carrying a handgun. Those pathways are closer to lethal-means access and policy screening than to the deterrence theory that dominated earlier right-to-carry debates. For that reason, this study treats firearm suicide as the primary outcome and firearm homicide as a separate secondary contrast.

The main empirical challenge is selection into adoption. States that adopted permitless carry tended to have higher baseline gun ownership, greater rurality, higher Republican presidential vote share, and higher baseline firearm-suicide rates than states that did not adopt. These differences do not preclude panel analysis, but they make interpretation demanding. A post-adoption coefficient can reflect the policy, pre-existing state trajectories, other firearm laws, political selection, health-access differences, substance-use environments, poverty, age structure, or mental-health workforce access.

Legal coding is a second source of uncertainty. Permitless open carry is not the same as permitless concealed carry, and several states do not fit cleanly into a binary adoption map. Vermont entered the panel already permitless, so it is not a within-panel adoption event. Arkansas has competing legal interpretations before and after its 2023 statutory clarification. Mississippi’s 2015 and 2016 changes differ in how broadly permitless concealed carry applied. Ignoring these distinctions can produce a precise estimate attached to an imprecise treatment definition.

We address these concerns in two ways. First, we use a source-audited legal treatment file that distinguishes clean within-panel adopters, verified non-adopters, baseline-permitless states, partial cases, and ambiguous reviewed cases. The audit also records whether adoption removed training requirements, carry-permit background checks, or permit-specific misdemeanor screens. Second, we prespecify an interpretive structure: firearm suicide is the primary outcome; total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks; and mechanism-oriented subgroup analyses are exploratory. State-specific trends, staggered-adoption estimators, Arkansas recoding, leave-one-out estimates, placebo timing, and covariate-layer sensitivity models are used to characterize robustness and limits rather than to create multiple independent headline findings.

The resulting claim is deliberately narrow. Permitless carry adoption was most consistently associated with higher firearm suicide rates. Total suicide also increased in baseline models, so the pattern is not a simple firearm-method-substitution result. Firearm homicide did not show a comparable positive association. The suicide estimates remained positive across many checks, but they attenuated under state-specific linear trends and short-window mental-health-provider

specifications, and event-time analyses showed pre-adoption signal. The contribution is therefore not a definitive causal estimate; it is a source-audited state-panel analysis showing where the firearm-suicide association appears, where it persists, and where it weakens.

Data and Treatment Coding

State-year panel

The analysis uses an annual 50-state panel from 1999 through 2024. Mortality outcomes come from CDC WONDER underlying-cause mortality files and are expressed as crude rates per 100,000 residents [12]. CDC WISQARS documentation provides the broader injury-surveillance context for fatal injury data derived from National Center for Health Statistics mortality files [13]. The primary outcome is firearm suicide. Secondary outcomes are total suicide, non-firearm suicide, firearm homicide, and total firearm deaths. Suicide and total-firearm models use 1,300 state-year observations; firearm-homicide models use 1,199 observations because some state-year homicide rates are unavailable or suppressed.

The baseline state-year covariates are unemployment and per-capita income. Sensitivity layers add external firearm-law indicators from the Tufts State Firearm Law Database [14], health-insurance access from Census SAHIE, drug-overdose mortality, Census demographic and poverty measures, and HRSA mental-health-provider access. RAND state-level household firearm-ownership estimates, rurality, presidential vote share, and baseline firearm-suicide rates are used for descriptive selection and heterogeneity analyses [15].

Treatment coding

The treatment is statewide permitless concealed carry: eligible adults can carry a concealed handgun in public without first obtaining a carry permit. The annual treatment indicator turns on in the first calendar year the statewide permitless concealed-carry rule is effective. Permitless open carry alone is not treated as permitless concealed carry.

The legal audit contains one row per state. It records 26 source-verified current-adopter rows, 21 verified non-adopter rows, one partial row, one ambiguous reviewed row, and one baseline-permitless row (Table 1). The processed panel contains 27 adoption-year states because the Mississippi coding is retained for consistency in the state-year treatment map; the audit separately marks Mississippi as partial and not clean policy-feature evidence. Arkansas is excluded from the primary clean-adoption map and is tested separately under 2021 and 2023 coding. Vermont is recorded as baseline permitless and is not treated as a within-panel adopter. The full state-by-state treatment table reports the state, adoption year, audit status, primary-map inclusion flag, coding note, and source URL trail (Supplementary Table S2).

Table 1: **Source-audited permitless-carry treatment map.**

Audit status	State count
Source-verified current adopter	26
Verified non-adopter	21
Ambiguous reviewed	1
Partial	1
Baseline permitless	1

The analytic panel retains 27 adoption-year states because Mississippi remains in the processed treatment map, while the legal audit marks it as partial. Arkansas is handled by sensitivity coding; Vermont is baseline permitless.

The audit also records carry-permit features for the 26 source-verified adopter rows. Twenty-one had a training requirement removed, 25 removed a carry-permit background-check screen, and 12 removed a permit-specific misdemeanor-violence screen. These fields describe the carry-permit process, not every underlying firearm-possession prohibition. We therefore treat them as descriptive policy features rather than as definitive mechanism tests.

Empirical Strategy

The baseline fixed-effects association model is a two-way fixed-effects state-year regression:

$$Y_{st} = \beta PostPermitless_{st} + \gamma_s + \delta_t + X_{st}\theta + \epsilon_{st},$$

where Y_{st} is an outcome rate in state s and year t , $PostPermitless_{st}$ is the post-adoption indicator, γ_s and δ_t are state and year fixed effects, and X_{st} includes unemployment and per-capita income in the baseline model. Standard errors are clustered by state. The coefficient β estimates the average post-adoption change associated with permitless carry after absorbing time-invariant state differences and national year shocks. The unweighted model is an average-state estimand; population-weighted models are reported separately because they answer the resident-weighted version of the same question.

This estimand is familiar in firearm-policy panel studies, but staggered adoption complicates interpretation when effects vary across cohorts or time [16–19]. We therefore report cohort-window ATT summaries, not-yet-treated dynamic ATT estimates, and never-treated-control event-time fixed-effect estimates. These analyses ask whether the direction of the result is similar under alternative timing comparisons and whether pre-adoption event-time patterns undermine a parallel-trends interpretation [20].

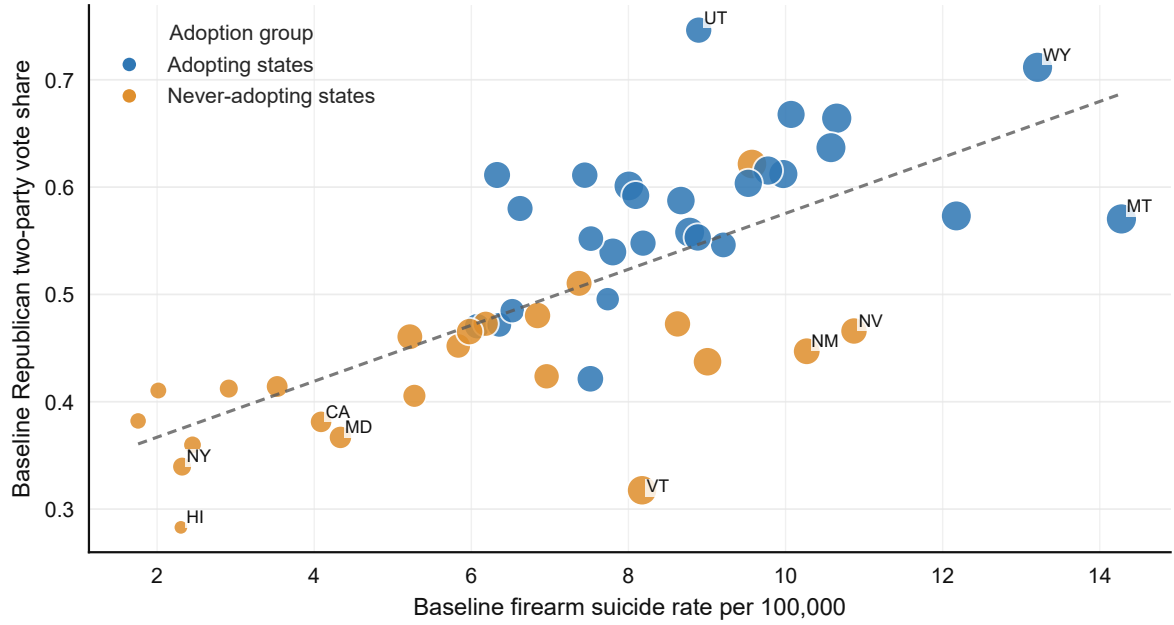
The outcome ordering was fixed before drafting. Firearm suicide is primary. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths assess outcome specificity. Sensitivity analyses add firearm-law covariates, non-firearm contextual covariates, demographic and mental-health covariates, state-specific trends, COVID-period exclusions, Arkansas recoding, leave-one-adopter-out checks, and placebo timing among never-treated states. We do not treat every covariate layer as a superior causal model because some time-varying covariates may reflect broader state policy environments. Heterogeneity and policy-feature analyses are hypothesis-generating because subgroup and feature-specific adopter counts are small.

Results

Adopting states differed before adoption

Before estimating mortality models, we compared adopter and non-adopter states on baseline state characteristics (Fig. 1). Adopter states had higher Republican two-party presidential vote share (0.579 vs. 0.425), higher estimated household gun ownership (0.479 vs. 0.342), greater rurality (0.683 vs. 0.461), and higher baseline firearm-suicide rates (8.847 vs. 5.734 deaths per 100,000). These differences make an unadjusted adopter-versus-non-adopter comparison uninformative.

Policy adoption is patterned by baseline risk and politics



Point size is proportional to baseline household firearm ownership. State-level baseline measures describe selection, not treatment effects.

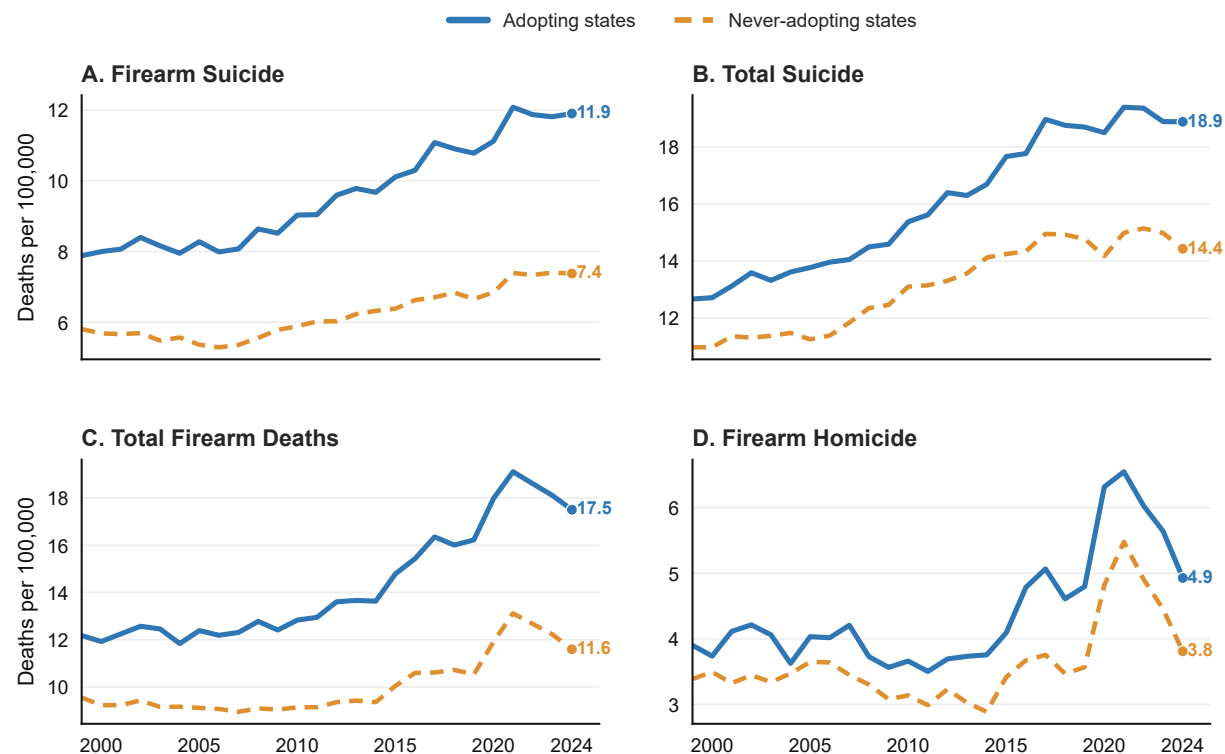
Figure 1: **Political and structural selection into permitless carry adoption.** The descriptive state-level comparison shows that adopter states differed from non-adopter states before or around adoption on political vote share, estimated gun ownership, rurality, and baseline firearm-suicide rates. The selection pattern motivates fixed effects, covariate-layer sensitivity checks, and staggered-adoption diagnostics.

The formal adoption logit was not stable because the small state sample and strong political sorting produced perfect-separation and convergence problems. We therefore use the descriptive comparisons as evidence of selection rather than interpreting unstable logit coefficients. This pattern frames the central empirical limitation: adopting and non-adopting states are not exchangeable without modeling assumptions.

Descriptive trends differed by outcome

Descriptive trends separated suicide and homicide before regression adjustment (Fig. 2). Firearm-suicide and total-suicide rates were higher in adopter states, while homicide trends were noisier and did not visually track the same post-adoption pattern. These trends are not causal evidence because states differed before adoption, but they motivate treating firearm suicide and firearm homicide as distinct endpoints.

Outcome trends by permitless carry adoption status



Annual state means, 1999-2024. Groups are defined by whether a state ever adopts permitless carry during the panel.

Figure 2: **Outcome trends by permitless-carry adoption status.** Descriptive trends motivate separate suicide and homicide analyses. They are not interpreted as causal estimates because adopter and non-adopter states differ structurally and politically before adoption.

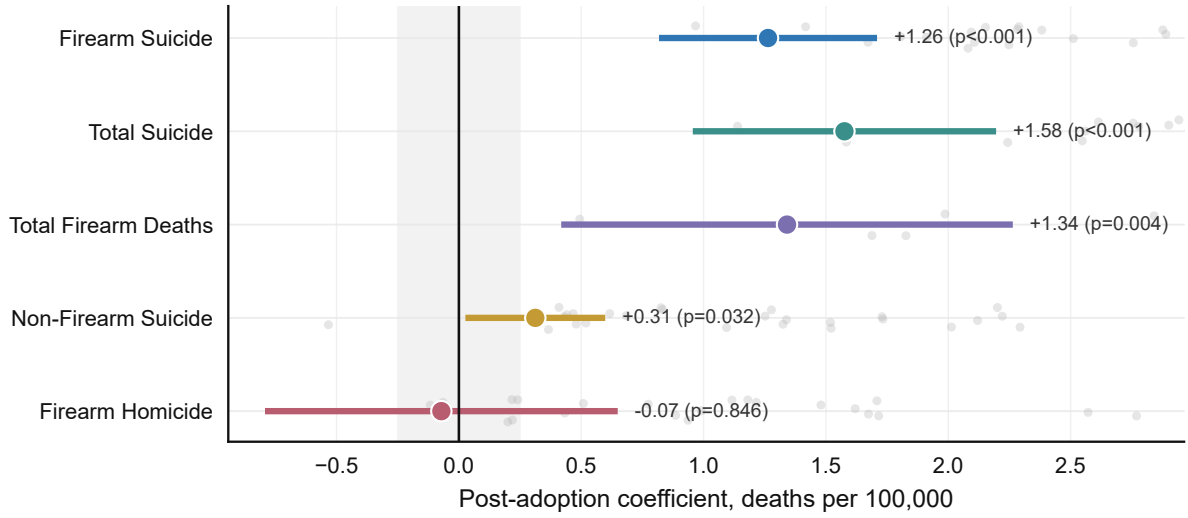
Baseline models showed a suicide association

The baseline fixed-effects model estimated a positive post-adoption association for firearm suicide (Table 2; Fig. 3). Permitless carry adoption was associated with a 1.263 deaths per 100,000 increase in firearm suicide rates (SE 0.227; $p < 0.001$). Total suicide also increased by 1.576 deaths per 100,000 (SE 0.316; $p < 0.001$), and total firearm deaths increased by 1.341 deaths per 100,000 (SE 0.471; $p = 0.004$). Non-firearm suicide was positive but smaller (0.312; SE 0.146; $p = 0.032$).

Table 2: **Baseline fixed-effects mortality estimates.**

Outcome	Coef.	SE	<i>p</i>	N
Firearm suicide	1.263	0.227	< 0.001	1300
Non-firearm suicide	0.312	0.146	0.032	1300
Total suicide	1.576	0.316	< 0.001	1300
Firearm homicide	-0.072	0.368	0.846	1199
Total firearm deaths	1.341	0.471	0.004	1300

Adjusted difference-in-differences estimates



Colored estimates are adjusted TWFE coefficients with +/- 1.96 SE intervals; pale points show unadjusted adopter-state pre/post changes.

Figure 3: **Baseline fixed-effects coefficient forest.** Models include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. Firearm suicide, total suicide, and total firearm deaths are positive in the baseline specification; firearm homicide is not.

Table 3 places the firearm-suicide estimate beside the main sensitivity checks. The association is positive across many specifications, including population weighting, pre-COVID restriction, COVID-period exclusion, and external firearm-law covariates. The most important attenuation occurs with state-specific linear trends, where the estimate falls to 0.253 ($p = 0.069$). Event-time fixed effects also show a positive post-adoption mean but a statistically significant pre-adoption signal. These results support the direction of the baseline association while limiting a strong causal interpretation.

Table 3: **Firearm-suicide diagnostic spine.**

Diagnostic row	Estimate	p	Interpretation
Baseline unweighted fixed effects	1.263	< 0.001	Average state-level association
Population-weighted fixed effects	0.853	< 0.001	Resident-weighted association
Pre-COVID only	1.137	< 0.001	Drops post-2019 years
COVID-excluded	1.238	< 0.001	Drops 2020–2021
External firearm-law covariates	0.994	< 0.001	Policy-environment sensitivity
State-specific linear trends	0.253	0.069	Central attenuation caveat
Cohort ATT windows	0.591/0.553/0.627	–	2-, 3-, and 5-year windows
Not-yet-treated dynamic ATT	0.431	–	Positive post-adoption mean
Event-time fixed effects	0.617	0.000007	Positive post mean; pretrend flagged

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Modern staggered-adoption rows are reported as diagnostics because event-time checks flag pre-adoption signal.

Firearm homicide did not follow the same pattern. The baseline fixed-effects estimate was -0.072 deaths per 100,000 (SE 0.368; $p = 0.846$), with 1,199 state-year observations. The estimate was statistically weak and centered near zero. Missingness remains relevant: 101 of 1,300 possible state-years were missing or suppressed, all in the smallest-population quartile (Table 4). This limitation tempers any strong homicide-null claim, but it does not account for the suicide-homicide contrast because total suicide and total firearm deaths show positive associations where firearm homicide does not.

Table 4: **Firearm-homicide missingness audit.**

Group	Possible	Observed	Missing	Missing pct.
All state-years	1300	1199	101	7.8%
Adopter-state state-years	702	636	66	9.4%
Non-adopter-state state-years	598	563	35	5.9%
Q1 smallest	325	224	101	31.1%
Q2	325	325	0	0.0%
Q3	325	325	0	0.0%
Q4 largest	325	325	0	0.0%

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

The main result is therefore outcome-specific: permitless carry is most consistently associated with suicide-related mortality, while firearm homicide shows no comparable positive association in the available state-year data.

Change-score windows showed the same outcome contrast

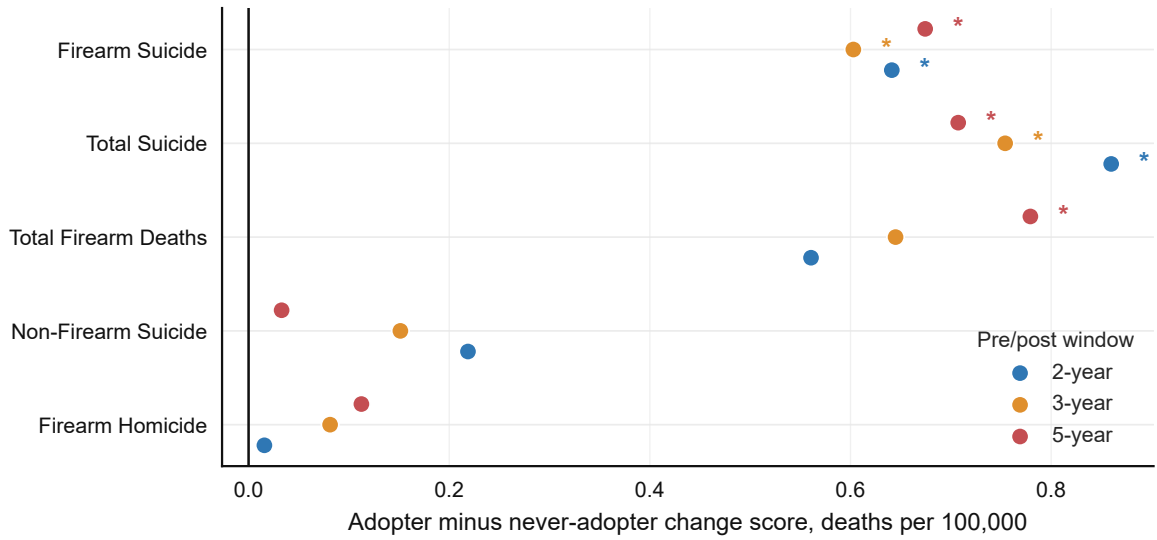
The pre-post change-score design asks a simpler question: within fixed windows around adoption, did adopter states experience larger changes than never-adopter states? Firearm suicide was positive and statistically significant in all three windows. The adopter-minus-never difference was 0.641 deaths per 100,000 in the 2-year window ($p = 0.001$), 0.603 in the 3-year window ($p < 0.001$), and 0.674 in the 5-year window ($p < 0.001$).

194 Firearm homicide again showed little evidence of a post-adoption increase (Table 5; Fig. 4). The
 195 corresponding homicide differences were 0.016 ($p = 0.944$), 0.081 ($p = 0.757$), and 0.112 ($p = 0.686$).
 196 Total suicide was positive in all three windows, while total firearm deaths became significant only in
 197 the 5-year window. Although less adjusted than the fixed-effects model, the change-score comparison
 198 shows that the suicide-homicide contrast is not unique to the baseline regression specification.

Table 5: **Pre-post change-score window checks.**

Outcome	2-year diff.	3-year diff.	5-year diff.
Firearm suicide	0.641**	0.603***	0.674***
Non-firearm suicide	0.219	0.151	0.033
Total suicide	0.860**	0.754***	0.707**
Firearm homicide	0.016	0.081	0.112
Total firearm deaths	0.561	0.645	0.779*

Change-score comparisons across robustness windows



Positive values indicate larger pre/post increases in adopting states. Asterisks denote Welch-test $p < 0.05$.

Figure 4: **Change-score robustness across pre-post windows.** Firearm suicide is positive across 2-, 3-, and 5-year windows. Firearm homicide remains statistically weak across the same windows.

199 Covariate layers attenuated the firearm-suicide estimate

200 Permitless carry may move alongside other firearm policies. We therefore added covariates for
 201 permit-to-purchase laws, waiting periods, universal background checks, ERPO/red-flag laws, safe-
 202 storage laws, stand-your-ground laws, and dealer licensing using the Tufts database [14]. The
 203 firearm-suicide estimate decreased from 1.263 to 0.994 deaths per 100,000 but remained statistically
 204 significant ($p < 0.001$). Total suicide and total firearm deaths also remained positive; firearm
 205 homicide remained statistically weak.

Suicide-related social environments may also differ across adopting and non-adopting states. Health-access covariates reduced the firearm-suicide estimate to 0.668 ($p < 0.001$), overdose covariates reduced it to 0.516 ($p = 0.044$), and demographic-poverty covariates reduced it to 0.648 ($p < 0.001$). These results suggest that the firearm-suicide association is not explained entirely by these measured domains.

The strongest attenuation came from specifications that included mental-health-provider access (Table 6). The HRSA mental-health-provider window is short, leaving only 150 state-year observations for the mental-health and full Phase 3B2 models. In that shorter window, firearm suicide remained positive but was no longer statistically significant (0.159, $p = 0.625$ for mental-health access; 0.197, $p = 0.546$ for the combined Phase 3B2 specification). These rows should not replace the full-panel model, but they identify an important boundary: the association weakens in a short recent window after adding mental-health-provider access and the full Phase 3B2 covariate set.

These models are best interpreted as sensitivity analyses, not as proof that adding more time-varying covariates necessarily improves causal identification. Some covariates, including overdose mortality, health access, poverty, and mental-health-provider access, may themselves reflect broader state policy environments or shocks related to adoption.

Table 6: **Firearm-suicide estimate across covariate-layer sensitivities.**

Specification	Coef.	p	N
Baseline fixed effects	1.263	< 0.001	1300
External firearm-law covariates	0.994	< 0.001	1300
Health-access covariates	0.668	< 0.001	800
Overdose covariates	0.516	0.044	300
Demographic-poverty covariates	0.648	< 0.001	1000
Mental-health access covariates	0.159	0.625	150
Full Phase 3B2 covariates	0.197	0.546	150

Staggered-adoption analyses raised pretrend concerns

Because adoption is staggered across states, we next examined cohort-window and event-time estimates (Table 7). Cohort-window ATT estimates for firearm suicide were positive across the 2-, 3-, and 5-year windows: 0.591, 0.553, and 0.627 deaths per 100,000. The not-yet-treated dynamic ATT summary also had a positive post-adoption mean for firearm suicide (0.431), while firearm homicide had a negative post-adoption mean (-0.030).

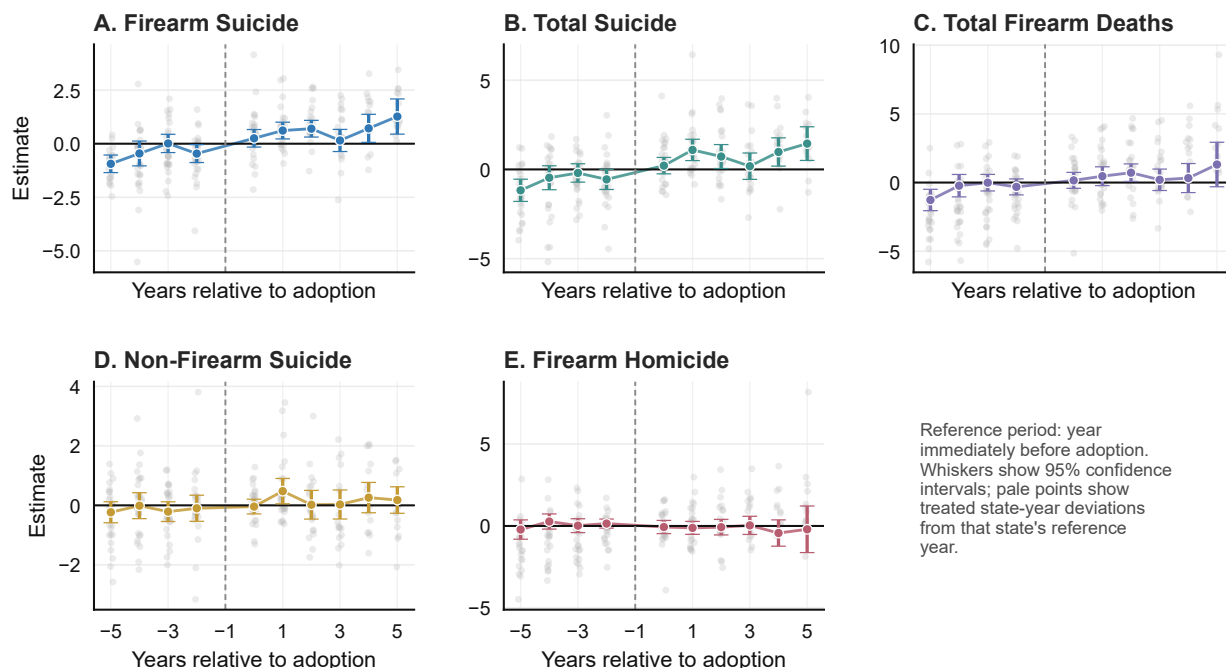
Table 7: **Staggered-adoption diagnostic summary.**

Outcome	ATT w2	ATT w3	ATT w5	Not-yet post	Interpretation
Firearm suicide	0.591	0.553	0.627	0.431	Positive, pretrend flag
Non-firearm suicide	0.263	0.232	0.199	0.050	Positive diagnostic
Total suicide	0.854	0.785	0.827	0.481	Positive, pretrend flag
Firearm homicide	0.085	-0.011	0.037	-0.030	No positive diagnostic
Total firearm deaths	0.391	0.454	0.557	0.360	Positive, pretrend flag

The event-time estimates are more cautionary (Fig. 5). Firearm suicide had positive post-adoption coefficients, but it also had statistically significant pre-adoption coefficients in the never-treated-

control event-time model. Total suicide and total firearm deaths showed similar pretrend patterns. Firearm homicide had no positive post-adoption pattern. These results are consistent with the direction of the suicide association but do not support a clean parallel-trends interpretation.

Event-study estimates around adoption



Dynamic coefficients are plotted relative to the year before permitless carry adoption; background points show the underlying treated-state event-time data.

Figure 5: **Event-time diagnostic estimates.** Event-time models show post-adoption suicide-related increases but also pre-adoption signals for firearm suicide, total suicide, and total firearm deaths. Those pretrend flags prevent the staggered-adoption results from serving as confirmatory causal estimates.

Robustness checks identified trend sensitivity

The firearm-suicide estimate remained positive under COVID-period exclusion (1.238; $p < 0.001$), pre-COVID restriction (1.137; $p < 0.001$), population weighting (0.853; $p < 0.001$), and external firearm-law covariates (0.994; $p < 0.001$). Leave-one-adopter-out estimates ranged from 1.135 to 1.356, and the observed firearm-suicide coefficient exceeded the 95th percentile of the never-treated placebo-timing distribution.

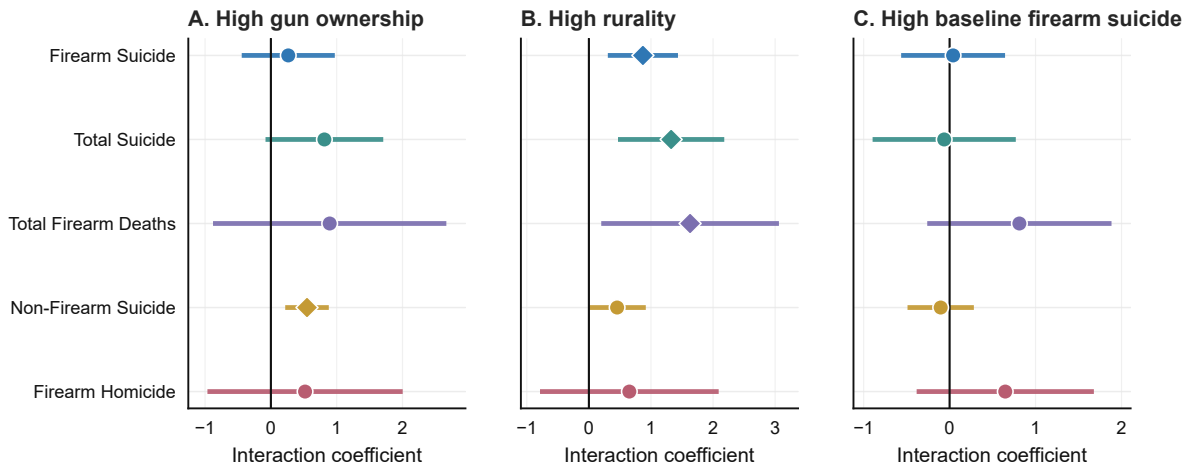
State-specific linear trends were the main exception. With state trends, the firearm-suicide estimate attenuated to 0.253 and was not statistically significant at the 0.05 level ($p = 0.069$). Total firearm deaths also attenuated strongly (0.117; $p = 0.664$). Total suicide remained positive (0.343; $p = 0.047$). These rows show that part of the firearm-suicide association is difficult to distinguish from longer state-specific trajectories.

Arkansas recoding did not change the broad pattern. When Arkansas was coded as treated in 2021 or 2023, firearm suicide remained positive and significant (1.261 and 1.277), total suicide remained positive and significant (1.595 and 1.601), and firearm homicide remained statistically weak (-0.031 and -0.066). The main outcome contrast is therefore not driven by the Arkansas coding decision.

Heterogeneity patterns were strongest for rurality

The clearest heterogeneity pattern was rurality (Fig. 6). Firearm suicide had a positive high-rurality interaction (0.870; $p = 0.003$), total suicide also had a positive high-rurality interaction (1.325; $p = 0.002$), and total firearm deaths had a positive high-rurality interaction (1.630; $p = 0.026$). Non-firearm suicide was more strongly associated with high estimated gun ownership (0.549; $p = 0.001$). Firearm homicide heterogeneity estimates were not statistically strong.

Heterogeneity in post-adoption associations



Interaction terms estimate whether associations are larger in states above each baseline median; diamonds mark $p < 0.05$.

Figure 6: Exploratory heterogeneity checks. Rurality is the clearest recurring heterogeneity dimension for firearm suicide, total suicide, and total firearm deaths. These models are exploratory because subgroup indicators are coarse and adoption is politically patterned.

The data do not directly observe carrying frequency, firearm acquisition, firearm storage, crisis timing, impulsive crises, or household firearm access. Plausible pathways include reduced pre-carry screening, reduced delay, weaker training requirements, and normalization of firearm availability, but the state-year panel cannot distinguish among them. The mortality pattern is therefore consistent with higher suicide-related risk after adoption, but it does not identify a specific firearm-access mechanism.

The policy-feature models did not identify a stable mechanism. None of the 15 training, permit-background-check, or misdemeanor-screen interaction rows had $p < 0.05$. The permit-background-check interaction could not be estimated because all 26 source-verified adopter rows removed the carry-permit background-check screen or an equivalent carry-permit screen, leaving no source-verified comparison group. Training removal and misdemeanor-screen removal had estimable rows, but the interaction estimates were not statistically strong. These results are best read as descriptive feature checks rather than evidence that one permit component explains the mortality association.

Discussion

This study evaluates a specific, recent form of concealed-carry deregulation: the removal of the carry-permit requirement. It does not estimate the full history of right-to-carry laws, and it does

not test whether public carrying deters crime. In a 50-state panel with source-audited treatment coding, the clearest association was between permitless carry adoption and firearm suicide.

The suicide result was not confined to a single specification. Firearm suicide increased in the baseline fixed-effects model, in 2-, 3-, and 5-year change-score windows, after external firearm-law covariates, after health-access covariates, after overdose covariates, after demographic-poverty covariates, in leave-one-adopter-out checks, under Arkansas recoding, and in several staggered-adoption summaries. The estimate also remained positive when population-weighted. Total suicide and total firearm deaths often moved in the same direction. Firearm homicide did not. This contrast is the central empirical pattern in the paper: permitless carry adoption is more consistently associated with suicide-related mortality than with homicide in this state-year panel.

The pattern is not a clean method-substitution result. If permitless carry only shifted suicide method toward firearms, firearm suicide would rise while total suicide remained stable or non-firearm suicide fell. Instead, total suicide increased in the baseline fixed-effects model, and non-firearm suicide was positive in that same specification. The non-firearm and total-suicide results were less stable across sensitivity layers, so firearm suicide remains the most durable signal. Still, the broader suicide pattern argues against describing the finding as simple substitution across methods.

The main limitation is identification. State-specific linear trends attenuated the firearm-suicide estimate, and event-time models showed pre-adoption signals for firearm suicide, total suicide, and total firearm deaths. These patterns mean the estimates should not be presented as proof that permitless carry caused the observed suicide increases. They suggest that adopting states were already on different mortality trajectories, or that policy timing coincided with other changes not fully captured by the covariates. The contribution is to document a consistent association and its limits, not to eliminate all remaining concerns about confounding.

The legal audit is an important part of the study because treatment coding is itself a source of uncertainty. Arkansas, Mississippi, and Vermont illustrate the problem. A state can be baseline permitless, partially permitless, ambiguously interpreted, or newly clarified by statute. A state can also remove the pre-carry permit application while retaining underlying possession prohibitions. The audit makes those distinctions visible and prevents the treatment variable from appearing more certain than the law allows.

The feature analysis should also be interpreted cautiously. Training removal, background-check removal, and misdemeanor-screen removal are plausible pathways, and prior concealed-carry work has emphasized the importance of permit provisions [2]. In this dataset, however, feature groups are small or nearly universal among clean adopters. The absence of statistically significant feature interactions is therefore not strong evidence against those pathways; it more likely reflects limited variation in the current state panel.

These findings fit a broader firearm-policy literature that increasingly treats suicide and homicide as distinct outcomes. State firearm laws associated with lower homicide are not always the same as those associated with lower suicide [21, 22]. Recent suicide-specific evidence links handgun permits, waiting periods, and concealed-carry permit requirements to lower firearm suicide rates [7]. The present analysis extends that logic to the recent permitless-carry wave and finds that the clearest mortality association is again on the suicide side.

Several limitations remain. The panel is state-level and cannot identify individual behavior, carrying frequency, firearm acquisition, storage, mental-health crises, crisis timing, or interpersonal circumstances. Some homicide observations are missing because of suppression or incomplete availability, with missingness concentrated in small-population state-years. The HRSA mental-health-provider specifications use a short recent window and should not be treated as full-panel replacements. The policy-feature fields require deeper statute-level auditing before they can support mechanism claims. Finally, adoption is politically and structurally patterned, so unmeasured state-

specific trajectories remain a serious concern even after fixed effects and covariate-layer sensitivity checks.

The practical implication is primarily methodological. Permitless-carry studies should separate primary outcomes from secondary and exploratory checks, audit legal treatment timing, report edge-case coding, report both unweighted and population-weighted estimands, and use staggered-adoption diagnostics without treating them as a complete solution to identification problems. Under that standard, the strongest statement from this analysis is cautious but policy-relevant: states adopting permitless concealed carry experienced larger post-adoption increases in firearm suicide rates, while firearm homicide showed no comparably consistent evidence of change in the available data.

Additional Methods Details

All baseline fixed-effects models used ordinary least squares with state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. Robustness specifications excluded 2020–2021, restricted the panel to pre-2020 years, applied population weights, added state-specific linear trends, or added external firearm-law covariates. Covariate-layer sensitivity models added firearm-law domains, uninsured rates, overdose mortality, age structure, race and ethnicity shares, poverty rates, and mental-health-provider access as described in the repository output tables. The unweighted estimates target the average state-level association; the population-weighted estimates target the average resident-level association.

Change-score analyses computed, for each adopter state, the difference between the average post-adoption rate and the average pre-adoption rate in 2-, 3-, and 5-year windows. Never-adopter states were assigned comparable window summaries and compared with Welch tests. Staggered-adoption checks included cohort-window ATT estimates, not-yet-treated dynamic ATT estimates, and never-treated-control event-time fixed-effect estimates. These checks were designed as sensitivity analyses because several outcomes showed pre-adoption event-time signals.

The outcome order was fixed before drafting the paper. Firearm suicide is the primary outcome. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks. Firearm-law covariates, contextual covariate layers, staggered-adoption estimators, Arkansas recoding, leave-one-out influence, and placebo timing are sensitivity analyses. Rurality, gun ownership, baseline-risk heterogeneity, and policy-feature interactions are exploratory.

Data and Code Availability

Processed analysis outputs, source-audited policy coding files, figure-generation scripts, and the manuscript-ready result tables are maintained in the project repository. Raw CDC mortality files and firearm-law source files are governed by their originating agencies and are documented in the repository source appendix.

Author Contributions

B.K. and Y.W. assembled the policy audit, implemented the mortality panel analyses, generated figures and tables, interpreted the results, and drafted the manuscript.

Competing Interests

The authors declare no competing interests.

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